

# Chapter 8.1

## Reviews

# Chapter 8 - Reviews

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  - **Participants**
  - **Preparations**
  - **The FDR session**
  - **Post-review activities**
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- **Comparison of peer reviews methods**
- **Expert opinions**

# Introduction

- The **design document** is key. Remember, your design is the **thoughtful solution** to satisfying the requirements.
- It is checked **repeatedly** in the development process.
- Typically, **reviewed many times** before getting a stamp of approval to proceed with development.
- Unfortunately, we often don't find our own errors and thus we need others for reviews.
- Different **stakeholders** with different viewpoints are used in the review process.
- Business analysts, end-users, developers, tester, customer service, quality assurance, account managers cost center managers...

# Introduction

- A **review process** is : “a process or meeting during which a work product or set of work products is presented to project personnel, managers, users, customers, or other interested parties for comment or approval.” (IEEE)
- Essential to detect / correct errors in these **earlier** work products because the **cost of errors downstream is very expensive!!**
- Review Choices:
  - Formal Design Reviews (FDR)
  - Peer reviews (inspections and walkthroughs)
    - Used especially in design and coding phase
  - Many are accommodated at different levels with different levels of interest.
  - Very interesting, essential, and oftentimes very frustrating.

# Formal Design Reviews (FDR)

## **Direct objectives – Deal with the current project**

- a. To detect analysis and design errors as well as subjects where corrections, changes and completions are required      Analysis is usually where we provide a logical understanding / solution to the requirements. This is what the 'requirements analysis' is
- b. To identify new risks likely to affect the project. Done early and often.
- c. To locate deviations from templates, style procedures and conventions.
- d. To approve the analysis or design product. Approval allows the team to continue on to the next development phase.

## **Indirect objectives – are more general in nature.**

- a. To provide an informal meeting place for exchange of professional knowledge about methods, tools and techniques.

- b. To record analysis and design errors that will serve as a basis for future corrective actions. (very important)

## Review objectives

- Many different kinds of reviews that apply to different objectives.
- Reviews are not randomly thrown together.
- Well-planned and orchestrated.
  - Objectives, roles, actions, participation, .... Very involved tasks.
  - Generally, there are defined procedures, roles, and responsibilities.
- **Participants are expected to contribute in their area of expertise.**
- Idea behind reviews is to **discover problems NOT** to fix them/
- Typically fixed after review and ‘offline’ so to speak.
- **Very common to review design documents. Probably the most common review type.**
- Thus they are **usually well-prepared** initially prior to review.

# Formal Design Reviews

|                     |                                                  |
|---------------------|--------------------------------------------------|
| <b>DPR –</b>        | <b>Development Plan Review</b>                   |
| <b>SRSR –</b>       | <b>Software Requirement Specification Review</b> |
| <b><i>PDR –</i></b> | <b><i>Preliminary Design Review</i></b>          |
| <b>DDR –</b>        | <b>Detailed Design Review</b>                    |
| <b>DBDR –</b>       | <b>Data Base Design Review</b>                   |
| <b><i>TPR –</i></b> | <b><i>Test Plan Review</i></b>                   |
| <b>STPR –</b>       | <b>Software Test Procedure Review</b>            |
| <b>VDR –</b>        | <b>Version Description Review</b>                |
| <b><i>OMR –</i></b> | <b><i>Operator Manual Review</i></b>             |
| <b>SMR –</b>        | <b>Support Manual Review</b>                     |
| <b>TRR –</b>        | <b>Test Readiness Review</b>                     |
| <b>PRR –</b>        | <b>Product Release Review</b>                    |
| <b><i>IPR –</i></b> | <b><i>Installation Plan Review</i></b>           |

**Important to note that a design review can take place any time an analysis or design document is produced, regardless whether that document is a requirement specification or an installation document.**

- Personal experience in:
  - Operators' Manual
  - Users' Manual
  - Program Maintenance Manual
  - Staff Users Manual
  - Installation / Conversion Plan
  - Test Plan(s)
  - Depend on size of project!

# Participants in the Review

## •Review Leader

- Needs to be **external** to the project team.
- Must have knowledge and experience in development of the **review type**
- Seniority at least as high as the project leader.
- Good relationships** with project leader and team
- Generally, this person really needs to know the project's material and should **NOT be the project leader.**
  - Would impact objectivity!
  - Would miss things entirely!

# Participants in the Review

## •Review Team

- Oftentimes needs to be from **senior members** of the team with other **senior members from other departments**.
  - Why?
  - Escalation?
- Team size should be 3-5 members.
- Too large, and we have too much coordination.
- This is a time for serious business – not lollygagging!

# Preparations for the Design Review

- **Participants in the review** include the

- Review leader,
- Review team, and
- Development team.

- **Review Leader:**

- appoint team members;
- establishes schedule, place, all types of coordination / resolution in membership
- distribute design documents, and more

- **Review Team preparation:**

- review document; list comments PRIOR to review session.
- For large design docs, leader may assign parts to individuals;
- Complete a checklist

- **Development Team preparations:**

- Prepare a short **presentation** of the document.
- Focus on main issues **rather** than describing the process.
- This **assumes** review team has read document and is familiar with project's outlines.

# The Design Review Session

- Start with short presentation (development team)

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- Comments by members (review team)
- Comments discussed to **determine required actions** team must perform (review and development teams)
- Decisions regarding design product – tells project's progress.
  - **Full approval** - Continue on to next phase (may have minor corrections)
  - **Partial approval**
    - Continue to next phase for some parts of project;
    - **Major action items needed for remainder of project.**
    - **Continuation of these parts granted only after satisfactory completion of action items**
      - Granted by review team member assigned to review completed actions or by the full review team or special review team or by some other forum
  - **Denial of Approval** – demands a repeat of Design Review
    - Project has major defects, critical defects

# The Design Review Report

- This is the Review Leader's **primary responsibility** following the review.
- Important to **perform corrections early** and minimize delays to project schedule.
- Report contains: (Sometimes called an ACTION LIST)
  - Summary of review discussions
  - Decision about project continuation
  - Full list of action items – corrections, changes, additions the project team must perform. For each item, anticipated completion date and responsible person is listed.
  - Name of review team member assigned for follow up.

# Follow up Process

- The ‘follow-up person’ may be the review team leader him/herself
- Required to verify each action item fixed as condition for allowing project to continue to next phase.
- Follow up must be fully documented to enable clarification of the corrections in the future, if any.

# Oftentimes Problems

- Sometimes entire parts of DR are often worthless due to
  - **inadequately prepared review team**, or
  - **intentional evasion** of a thorough review.
- Tipoff:
  - **Very short report** – limited to documented approval of the design and listing few, if any, defects
  - **Short report** approving continuation to next project phase in full
    - listing several **minor** defects **but no action items**.
  - A report listing several action items of varied severity **but no indication of follow-up** (no correction schedule, no documented activities, ...)

## Pressman's 13 golden guidelines for a successful Design Review

### Design Review Infrastructure

- Develop checklists for common types of design documents.
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**Train senior professionals to serve as a reservoir for Design Review teams.**

- Periodically analyze past Design Review effectiveness.
- Schedule the Design Reviews as part of the project plan.

## Pressman's 13 golden guidelines for successful DR - 2

### The Design Review Team

- **Review teams size should be limited, with 3–5 members being the optimum.**

## Pressman's 13 golden guidelines for successful DR - 3

### The Design Review Session

- **Discuss professional issues** in a constructive way refraining from personalizing the issues.
- **Keep** to the review **agenda**. So easy to get sidetracked. So easy to get personal!!
- **Focus on detection of defects** by verifying and validating the participants' comments.
  - Refrain from discussing possible **solutions**.
- **If disagreement about an error** - end the debate by noting the issue and shifting its discussion to another forum.
- **Properly document discussed comments**, and the results of their verification and validation.
- **Review session should not exceed two hours.**

## Pressman's 13 golden guidelines for successful DR - 4

### Post-Review Activities

- Prepare the review report, including the action items
- Establish follow-up to ensure the satisfactory performance of all the list of action items